INCH POUND
MIL-B-63957 (AR)
February 23, 1990

MILITARY SPECIFICATION

BOX ASSEMBLY. PNEUMATIC LOCK KEY

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This military specification covers one type of key box assembly used in the operation of the pneumatic lock incorporated in the Weapon Access Delay System program.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standarization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-8140
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PRODUCT DRAWINGS

U.S. ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER

9354275	Manifold "A" Sub-Assembly
9378140	Box Assembly, Key
9378142	Connector
9378146	Encoder/Tubing Assembly
9378147	Panel Sub Assembly

INSPECTION EQUIPMENT DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPEMENT AND ENGINEERING CENTER

9354313	Key, Test

(Copies of drawings, and publications required by manufacturers in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.4)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly. The key box assembly shall comply with all requirements specified on drawings (dwg.) 9378140, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

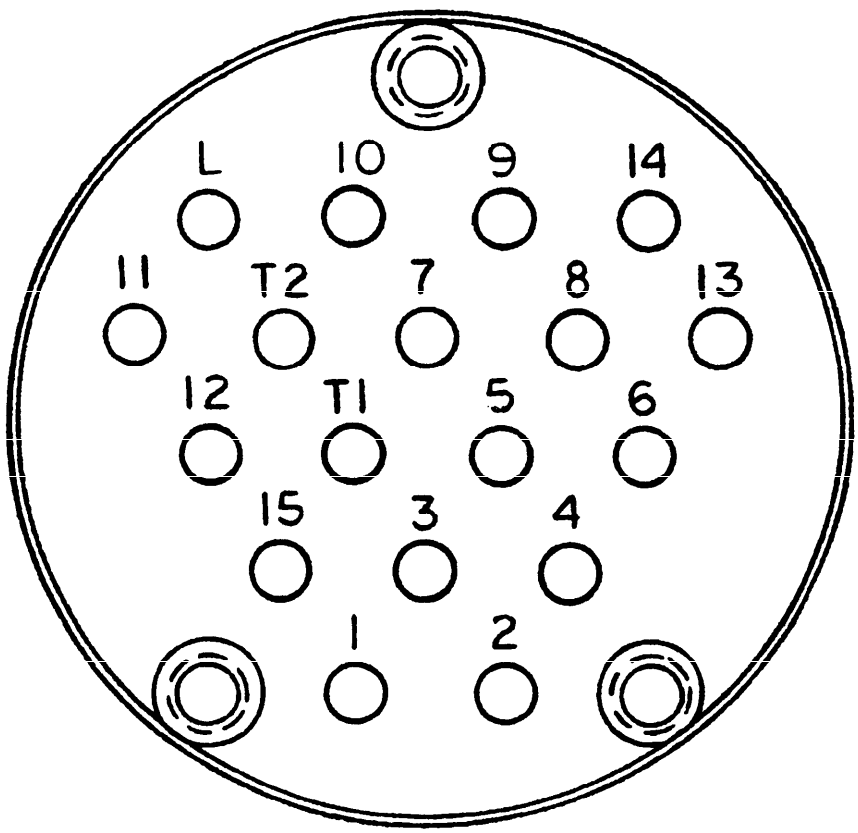
3.3 First article. When specified, a sample key box assembly shall be subjected to first article inspection (see 4.3 and 6.2).

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3.4 Function requirement.

3.4.1 Panel subassembly. The Panel Subassembly (9378147) of the Key Box Assembly (9378140) shall maintain a regulated nitrogen gas pressure of 100 ± 3 psig for a minimum duration of 3 minutes. When the needle valve is in the closed position, the maximum pressure drop shall be 1/4 psig. When the needle valve is in the open position, the maximum pressure drop shall be 1 psig. Testing shall be as specified in 4.5.1.1.

3.4.2 Encoder/Tubing Assembly. Each air path in the Encoder/Tubing Assembly (9378146) shown in Table I shall demonstrate an air pressure continuity. Input pressure shall be 70 psig maximum. Output pressure on ports 14 and 10 and T-Fitting and L-Fitting shall be 60 psig minimum. Output pressure on ports 1 thru 9 shall be 2 psig minimum. Nitrogen gas inputs and outputs to be through the face of connector 9378142 of the Encoder/Tubing Assembly as shown below. Testing shall be as specified in 4.5.1.2.



CONNECTOR - 9378142

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TABLE I. Function requirement

AIR INPUT (Connector 9378142)	TEST CONDITION	AIR OUTPUT
13 11 L T1 T2	Press Read Press Lock --------- Plug T2 Plug T1	14 10 L-Fitting T-Fitting T-Fitting
	Insert Key, Test 9354313-1 thru 9 Press Read	Connector (9378142)
12 and 15 Simultaneously	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9

3.5 Serialization of the panel subassembly. A serial number shall be assigned by the contractor to each panel subassembly produced (see dwg. 9378147). In no event shall serial numbers be duplicated, nor shall they give any indication of classified production quantities.

3.6 Workmanship. All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All pneumatic hoses must be free of kinks.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspectons set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. Where the contract requires first article inspection, the contractor shall submit two consecutively produced key box assemblies to the testing facility designated in the invitation for bids or request for proposal.

4.3.2 Inspection to be performed. The first article inspection samples shall be inspected for the defects specified in 4.4.2.1, using inspection methods contained therein, and for all the requirements of Section 3, as applicable. The key box assembly test requirements of Section 3 are listed in Table II, and the tests shall be performed in the listed sequence. The functional tests of the panel subassembly, (see 3.4.1) and the encoder/tubing assembly, (see 3.4.2) shall be performed before assembling into the key box assembly. Each item shall be tested with the methods of inspection specified in 4.5. Inspection

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shall be performed with the acceptance inspection equipment specified in 4.4.4. Should any item fail to comply with any of the applicable requirements, the first article samples shall be rejected. The Government reserves the right to terminate its inspections upon any failure of the first article samples to comply with the stated requirements. This series of tests, listed in Table II, is not considered destructive.

TABLE II. First Article Tests

Test	Requirement paragraph
a. Function Tests	3.4
1. Panel subassembly	3.4.1
2. Encoder/Tubing Assembly	3.4.2

4.4 Quality conformance inspection.

4.4.1 Inspection lot. A lot or batch and its formation, size and presentation is described in MIL-A-48078. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination. A sample shall be randomly selected from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table III). One defect found in the sample, selected for either the major or the minor inspections for the classification, shall reject the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspection are performed. The term, "SMTE," where used, shall mean any standard measuring and test equipment suitable for the test being performed. The term "TMC," where used, shall mean tool manufacture control (see 6.5).

TABLE III. Inspection lot sampling.

Major defect table		Minor defect table	
Lot size	Sample size	Lot size	Sample size
1-5	all	1-5	all
6-10	5	6-10	4
11-15	10	11-15	8
16-25	15	16-25	11
26-35	22	26-35	14
36-50	26	36-50	16
51-70	30	51-70	18
71-100	33	71-100	19
101 and over	37	101 and over	20

4.4.2.1 Classification of defects. Critical, major, and minor defects are defined in MIL-A-48078.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

MIL-B-63957 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1.1	Manifold "A" Sub-assembly	1 OF 2	9354275
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major:				
101.	Component missing, damaged, incorrect or improperly assembled	Table III	3.2	Visual
102.	Location of holes (9) incorrect	Table III	3.2	SMTE
103.	Diameter of holes (9) incorrect	Table III	3.2	SMTE
104.	Location of holes (2 for pins) incorrect	Table III	3.2	SMTE
105.	Diameter of holes (2 for pins) incorrect	Table III	3.2	SMTE
106.	Location of flat bottom hole (2) incorrect	Table III	3.2	SMTE
107.	Diameter of flat bottom hole (2) incorrect	Table III	3.2	SMTE
108.	Depth of flat bottom hole (2) incorrect	Table III	3.2	SMTE
109.	Coaxiality of holes (9) not met	Table III	3.2	SMTE
110.	Alignment of manifolds incorrect	Table III	3.2	SMTE
111.	Relative movement in plane between manifolds greater than specified	Table III	3.2	SMTE
112.	Protective finish on excepted areas	Table III	3.2	Visual
113.	Key groove slot incorrect (.14 + .01) and (.76 + .01)	Table III	3.2	SMTE

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-B-63957(AR)

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.1.1	Manifold "A" Sub-assembly		9354275
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor:				
201.	Protective finish missing, incomplete or incorrect	Table III	3.2	Visual
202.	Evidence of poor workmanship	Table III	3.6	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-B-63957 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1.2	Connector	1 OF 1	9378142
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major:				
101.	Location of tapped holes (21) incorrect	Table III	3.2	SMTE
102.	Pitch diameter (Max) of tapped holes (21) incorrect	Table III	3.2	SMTE
103.	Minor diameter (Min) of tapped holes (21) incorrect	Table III	3.2	SMTE
104.	Diameter of countersink (3 places) incorrect	Table III	3.2	SMTE
105.	Depth of countersink (3 places) incorrect	Table III	3.2	SMTE
106.	Outside diameter incorrect	Table III	3.2	SMTE
107.	Flatness requirement not met	Table III	3.2	SMTE
Minor:				
201.	Chamfer in excess of requirements on drawing (18 places)	Table III	3.2	SMTE
202.	Evidence of poor workmanship	Table III	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1.3	Panel Subassembly	1 OF 1	9378147
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101.	Component missing, damaged, incorrect or improperly assembled	Table III	3.2	Visual
102.	Tape missing on pipe thread joints	Table III	3.2	Visual
103.	Torque on nut not between limits	Table III	3.2	SMTE
104.	Function test requirements not met	100%	3.4	TMC
Minor:				
201.	Serialization missing, improper or illegible	Table III	3.2	Visual
202.	Evidence of poor workmanship	Table III	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

MIL-B-63957 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1.4	Encoder/Tubing Assembly	1 OF 1	9378146
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101.	Component missing, damaged, incorrect or improperly assembled	Table III	3.2	Visual
102.	Length incorrect (30.00 ± 1.00)	Table III	3.2	SMTE
103.	Distance from far face of Encoder Assembly to near side of "L" Fitting incorrect	Table III	3.2	SMTE
104.	Function test requirements not met	100%	3.4	TMC
Minor:				
201.	Evidence or poor workmanship	Table III	3.6	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1.5	Box Assembly, Key	1 OF 1	9378140
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major:				
101.	Component missing, damaged, incorrect or improperly assembled	Table III	3.2	Visual
Minor:				
201.	Evidence of poor workmanship	Table III	3.6	Visual

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4.4.3 Quality conformance inspection testing. The contractor shall perform all of the tests specified in Table IV. The tests may be performed in any sequence.

TABLE IV. Quality conformance tests.

Test	Test classification
Function tests:	
1. Panel Subassembly	Major
2. Encoder/Tubing Assembly	Major

4.4.3.1 Function tests: (1.) panel subassembly (see 3.4.1) and (2) encoder/tubing assembly (see 3.4.2). Each panel subassembly, and encoder/tubing assembly shall be subjected to these tests, as applicable. Should any panel subassembly or encoder/tubing assembly fail to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests. Inspection shall be performed with the acceptance inspection equipment, operating instructions, and calibration procedures designed or specified by the contractor. The contractor shall obtain approval of such equipment designs, operating instructions, and calibration procedures prior to use on the contract (see 6.4).

4.4.4.2 Minor defects. The acceptance inspection equipment, operating instructions, and calibration procedures used by the contractor for minor classification inspection shall be approved by the Government representative responsible for acceptance inspection.

4.5 Methods of inspection. The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature (77°± 10°F).

4.5.1 Function tests. The pneumatic source to be used for all function tests shall be a supply of dry, filtered, compressed nitrogen gas.

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4.5.1.1 Panel subassembly. The panel subassembly shall be pressurized to 100 ± 3 psig utilizing an approved procedure and measuring test equipment.

The assembly shall be pressure stabilized with the tube cap tightened, the needle valve in the open position, the adapter of the test assembly engaged as the pressure input terminal, and an approved pressure gage attached to the braid hose of the test assembly.

After the test stabilization setup is completed, close the needle valve and seal off the nitrogen gas supply. The observed pressure drop shall be less than 1/4 psig during the subsequent three minutes (minimum) duration.

Repeat the test stabilization setup and with the needle valve kept in the open position, seal off the adapter. The observed pressure drop shall be less than one psig during the subsequent three minute (minimum) duration.

The panel subassembly shall comply with the requirements of 3.4.1.

4.5.1.2 Encoder/tubing assembly. The encoder/tubing assembly shall be pneumatically tested to determine that a given input causes an output as specified in 3.4.2. The test shall be performed utilizing an approved procedure and measuring equipment.

Nine coded keys (reference 9354313) shall interface with the encoder/tubing assembly to obtain the gas flow alignment as designated in Table I.

No audible gas leakage sound shall occur during the duration of this test.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. The box assembly shall be preserved, packaged, packed, and marked in accordance with P9378140 or as specified in the contract or purchase order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The box assemblies covered by this specification are intended for use in pneumatic lock components for the weapon access delay systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of control box assembly being procured, part number and nomenclature.

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. When first article is required (see 3.3).

e. Applicable methods of packaging and packing (see 5.1).

6.3 Objective evidence

6.3.1 General. Contractors inspection records and documents verifying compliance with applicable dwg. and specification requirements.

6.3.2 Materials. A statement (certification), supported by test data, that all materials produced or purchased by the contractor meets all requirements when such material is controlled by Government or commercial specification referenced in contractual documents.

6.4 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration procedures from Commander, U.S. Army Armament, Research, Development and Engineering Center, ATTN: AMSMC-QAN-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Tool manufacture control (TMC). When TMC is specified a a method of inspection, it shall mean that the manufacturing tool setup (e.g., stamping die, automatic tape control machine, etc.) may be used to accept the parts being manufactured. Two consecutive parts, which comply with drawing dimensions, shall constitute proper initial tool control acceptance. In addition, two consecutive parts shall be examined, as specified in the applicable paragraph in the classification of defects, whenever the manufacturing tool requires maintenance.

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6.6 Subject term (key word) listing.

Key box assembly	acceptance equipment
encoder	pneumatic logic
manifold	functional gage

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A806)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-63957 (AR)	23 February 1990

3. DOCUMENT TITLE

BOX ASSEMBLY, PNEUMATIC LOCK KEY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: (201) 724-6671 (2) AUTOVON: 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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